International Islamic University Chittagong

Department of Computer Science and Engineering

LAB Mid-Term Examination, Autumn 2025 (SET B)

Course: CSE-4875 (Pattern Recognition and Image Processing) Section: _____

Time: 1 Hour 30 Minutes **Marks: 30**

[Answer all questions; Figures in the right-hand margin indicate Marks]

Before you start. In the first cell of your notebook write your **name**, **student ID**, and the two digits below. Every question uses them.

- **a** = the last digit of your student ID
- **b** = the second-last digit of your student ID

Example: for ID C221047, a = 7 and b = 4.
In Questions 2 and 3 you must use nested `for` loops. Library functions that do the whole operation for you are not allowed there, except in part (d) of Question 3. Save the notebook with all outputs visible before you leave.

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1. Type the code below exactly, replacing `a` and `b` with your own digits. It builds your test image. Then read and inspect it. **[5]**

```
import numpy as np
import matplotlib.pyplot as plt

a = 0      # <-- last digit of YOUR student ID
b = 0      # <-- second-last digit of YOUR student ID

np.random.seed(10*b + a)
img = np.zeros((256, 256), np.uint8)
img[:] = (60 + 0.45*np.arange(256)).astype(np.uint8)
img[60:150, 70:180] = 150 + 5*a
img[180:230, 40:120] = 20 + 5*b
n = np.random.rand(256, 256)
img[n < 0.03] = 0
img[n > 0.97] = 255
```

(a) Print the height, width, `dtype`, minimum, maximum and mean of `img`.

(b) Display `img` correctly as a greyscale image.

(c) Crop rows **10a** to **10a + 100** and columns **10b** to **10b + 100**. Display the crop and save it as `crop_<your ID>.png`.

2. Perform the following grey level transformations on `img`. **Use nested `for` loops only.** Display the original and each result side by side. **[8]**

(a) **Min-max contrast stretching.** Use $s = (r - r_1) \times 255 / (r_2 - r_1)$, where r_1 is the minimum and r_2 the maximum grey level of the image. State the r_1 and r_2 you used. **[5]**

(b) **Intensity level slicing** of the original image. Take T = **100 + 5a**. Pixels whose value lies between T and T + 60 inclusive become 255, all others become 0. State your T. **[3]**

3. Apply a **3 × 3 median filter** to `img`. **[12]**

(a) Pad the image with a one-pixel border of zeros. **[2]**

(b) Using nested `for` loops, take the 3 × 3 neighbourhood at each position, compute its median, and write it to the output image. **[7]**

(c) Display the original and your filtered result side by side. **[1]**

(d) Now apply `cv2.medianBlur(img, 3)` and display it too. Print the value of **your** output and of the OpenCV output at pixel **(50 + a, 50 + b)**. Write one sentence saying whether the two images agree everywhere. **[2]**

4. Viva at your seat. **[5]**

End of Set B